

Mingnan Yao

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Ann Arbor, MI 48105

Focus Area

- Biomedical Data Science; Applied Machine Learning; Multi-Modal Data Modeling

Education

- **University of Michigan** *Anticipated Graduation: Apr. 2027*
B.S. in Biomedical Engineering Ann Arbor, MI
 - GPA: 4.00/4.00
 - Current Coursework: Intro to Biomedical Engineering Design, Foundations of Computer Vision
- **Shanghai Jiao Tong University** *Anticipated Graduation: Aug. 2027*
B.S. in Electrical and Computer Engineering Shanghai, China
 - GPA: 3.89/4.00
 - Courses with Distinction: Data Structures and Algorithms, Electronic Circuits, Linear Algebra

Research

- **Research Assistant – Multi-Modal Single-Cell Deep Learning** *Sept. 2024 – Present*
University of Michigan
 - Reproduced and extended deep learning models for multi-modal single-cell analysis (RNA-seq + ATAC-seq) based on recent literature.
 - Worked with PyTorch-based research codebases to modify model components and training pipelines, improving performance on downstream biological prediction tasks.
- **Research Project – Dexterous Hand Grasping** *Sept. 2024 – Aug. 2025*
Shanghai Jiao Tong University
 - Conducted an extensive literature review on methods in dexterous robotic grasping, with emphasis on control strategies and learning-based approaches.
 - Synthesized insights from robotics and machine learning to identify key challenges and promising directions, laying the groundwork for future experimental studies.

Experience

- **Teaching Assistant – Introduction to Circuits** *May – Aug. 2025*
Shanghai Jiao Tong University
 - Led weekly recitation sessions for 90+ students and held office hours to clarify key circuit analysis concepts.
 - Guided students in laboratory sessions by providing step-by-step instruction on circuit design, measurement, and troubleshooting.

Activities

- **Member of Honor Council** *Mar. 2025 – Present*
SJTU
 - Contributed to fostering academic integrity by reviewing cases and promoting awareness initiatives within the Joint Institute community.
- **Director of Publicity** *Dec. 2023 – Aug. 2025*
Joint Institute Physiology Association (JIPA), SJTU
 - Oversaw publicity strategy and coordinated a team to promote events through posters, newsletters, and social media platforms, expanding student participation and visibility.

Skills

- **Computer Languages:** Python/PyTorch, C/C++, MATLAB
- **Tools:** Git, LaTeX, Google Workspace, SolidWorks
- **Languages:** English, Chinese

Honors & Awards

- National Scholarship (2025) | Cheng Family Scholarship (2025) | Yuze Scholarship (2024)